

Model Reduction of Constrained Mechanical Systems via Spectral Submanifolds: Application to a 1:2 Internally Resonant Hyperbolic Paraboloid

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July 17, 2026

Abstract

Dynamical systems in engineering are frequently subjected to algebraic configuration constraints in conjunction with their governing ordinary differential equations (ODEs), rendering them as differential-algebraic equations (DAEs). A full-scale numerical simulation of such high-dimensional constrained systems is computationally expensive and obfuscates the underlying nonlinear modal interactions. In this paper, we present a mathematically rigorous model reduction framework for constrained mechanical systems based on the theory of Spectral Submanifolds (SSMs). We seamlessly extend the SSM parameterization to descriptor-form DAEs without the need for explicit index-reduction algorithms. To demonstrate the necessity of correct subspace selection, we investigate a three-degree-of-freedom oscillator strictly confined to a hyperbolic paraboloid manifold and tuned to exhibit a 1:2 internal resonance. We formally show that classical two-dimensional invariant manifold theories fail in this regime, whereas a four-dimensional SSM perfectly captures the energy beating phenomena and the algebraic constraint forces. Furthermore, we extract the forced response curves (FRCs) and backbone surfaces analytically up to $\mathcal{O}(11)$, providing exact agreement with full-order numerical continuations.

1 Introduction

Decomposing nonlinear oscillations in analogy with linear modal analysis has been an ongoing pursuit in structural dynamics for several decades. In the engineering mechanics literature, this approach was initiated by Rosenberg [1], who defined a nonlinear normal mode (NNM) in a conservative system as a synchronous periodic oscillation. Shaw and Pierre [2] offered an elegant geometric alternative, envisioning NNMs as invariant two-dimensional manifolds in the phase space that are tangent to the linear modal subspaces at the equilibrium.

While the Shaw-Pierre invariant manifold approach provides a powerful conceptual tool, its application to general dissipative and forced systems is mathematically precarious. As pointed out by Haller and Ponsioen [4], in the presence of damping, the phase space contains an uncountably infinite number of invariant manifolds tangent to a given linear subspace. Approximating an arbitrary manifold from this infinite family via Taylor series often leads to divergent behavior and lack of uniqueness, a phenomenon already noted by Euler for simple planar systems.

To resolve this ambiguity, Haller and Ponsioen [4] introduced the theory of *Spectral Submanifolds (SSMs)*. An SSM is defined as the unique, smoothest nonlinear invariant manifold that serves as the continuation of a linear spectral subspace of the unforced system. By relying on classical results by Cabré, Fontich, and de la Llave [3], the SSM framework provides exact existence and uniqueness conditions based solely on the spectral quotients of the linearized system. The flow restricted to the SSM provides a mathematically exact Reduced-Order Model (ROM) of the full-scale system.

Recently, the SSM theory has been extended to systems subject to algebraic configuration constraints (e.g., multibody mechanisms, incompressible flows) by Li, Jain, and Haller [5]. In constrained systems, the governing equations take the form of index-3 Differential-Algebraic Equations (DAEs). The reduction of DAEs typically requires cumbersome coordinate partitioning or Baumgarte stabilization [6]. However, by casting the DAEs into a first-order descriptor form, the SSM parameterization can directly digest the constraints, yielding a low-dimensional ROM composed purely of Ordinary Differential Equations (ODEs) while inherently preserving the exact geometric constraints of the full phase space.

In this work, we build upon these foundational results to address a critical topology in nonlinear dynamics: *internal resonances* within constrained manifolds. When the natural frequencies of a system are commensurate (e.g., 1:2, 1:3), energy is continuously transferred between the interacting modes. We demonstrate that reducing such a system to a two-dimensional SSM results in catastrophic failure due to the violation of non-resonance conditions. Instead, an appropriately dimensioned multi-mode SSM must be constructed.

We illustrate this through a novel mechanical architecture: a generalized oscillator strictly constrained to a saddle surface (a hyperbolic paraboloid) operating under a 1:2 internal resonance. We provide the full DAE formulation, compute the high-order SSMs analytically, validate the transient energy exchange, and extract the high-fidelity Forced Response Curves (FRCs) compared against state-of-the-art numerical continuation via the `COCO` package [7].

2 Mechanical Systems with Configuration Constraints

2.1 Differential-Algebraic Equation (DAE) Formulation

We consider a periodically forced nonlinear mechanical system subject to n_c holonomic configuration constraints. The governing equations of motion, derived via the Euler-Lagrange formalism, are given by:

$$\begin{aligned} \mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} + \mathbf{f}_{nl}(\mathbf{x}, \dot{\mathbf{x}}) + \mathbf{G}(\mathbf{x})^\top \boldsymbol{\mu} &= \epsilon \mathbf{f}_{ext}(\mathbf{x}, \dot{\mathbf{x}}, \Omega t), \\ \mathbf{g}(\mathbf{x}) &= \mathbf{0}, \end{aligned} \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the displacement vector; $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{n \times n}$ are the mass, damping, and stiffness matrices respectively. The vector function $\mathbf{f}_{nl}(\mathbf{x}, \dot{\mathbf{x}}) = \mathcal{O}(|\mathbf{x}|^2, |\mathbf{x}||\dot{\mathbf{x}}|, |\dot{\mathbf{x}}|^2)$ encapsulates the smooth geometric and material nonlinearities. The term $\epsilon \mathbf{f}_{ext}$ represents external harmonic excitation with forcing frequency Ω , where $0 \leq \epsilon \ll 1$ is a small scaling parameter.

The vector $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^{n_c}$ ($n_c < n$) represents the holonomic configuration constraints, and $\mathbf{G}(\mathbf{x}) = \partial \mathbf{g} / \partial \mathbf{x} \in \mathbb{R}^{n_c \times n}$ is the full-rank Jacobian matrix of these constraints. The vector $\boldsymbol{\mu} \in \mathbb{R}^{n_c}$ contains the Lagrange multipliers, physically representing the magnitude of the constraint forces required to keep the trajectory on the manifold $\mathbf{g}(\mathbf{x}) = \mathbf{0}$.

Without loss of generality, we assume that $\mathbf{g}(\mathbf{0}) = \mathbf{0}$, making the origin $(\mathbf{x}, \dot{\mathbf{x}}, \boldsymbol{\mu}) = \mathbf{0}$ an equilibrium point of the unforced system ($\epsilon = 0$). By expanding the constraint function, we can express it as:

$$\mathbf{g}(\mathbf{x}) = \mathbf{G}_0 \mathbf{x} + \mathbf{g}_{nl}(\mathbf{x}) = \mathbf{0}, \quad (2)$$

where $\mathbf{G}_0 = \mathbf{G}(\mathbf{0})$ and $\mathbf{g}_{nl}(\mathbf{x}) = \mathcal{O}(|\mathbf{x}|^2)$.

To seamlessly apply the invariant manifold theory, we transform the second-order DAE system (1) into a first-order descriptor form. We define the augmented state vector $\mathbf{z} = (\mathbf{x}, \dot{\mathbf{x}}, \boldsymbol{\mu})^\top \in \mathbb{R}^{2n+n_c}$ and write:

$$\mathbf{B}\dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{F}(\mathbf{z}) + \epsilon \mathbf{F}_{ext}(\mathbf{z}, \Omega t), \quad (3)$$

where the global linear matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{(2n+n_c) \times (2n+n_c)}$ are given by

$$\mathbf{A} = \begin{bmatrix} -\mathbf{K} & \mathbf{0} & -\mathbf{G}_0^\top \\ \mathbf{0} & \mathbf{M} & \mathbf{0} \\ \mathbf{G}_0 & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{C} & \mathbf{M} & \mathbf{0} \\ \mathbf{M} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}. \quad (4)$$

The global nonlinear vector field $\mathbf{F}(\mathbf{z})$ and external forcing $\mathbf{F}_{ext}$ are constructed as:

$$\mathbf{F}(\mathbf{z}) = \begin{bmatrix} -\mathbf{f}_{nl}(\mathbf{x}, \dot{\mathbf{x}}) - (\mathbf{G}(\mathbf{x}) - \mathbf{G}_0)^\top \mu \\ \mathbf{0} \\ \mathbf{g}_{nl}(\mathbf{x}) \end{bmatrix}, \quad \mathbf{F}_{ext}(\mathbf{z}, \Omega t) = \begin{bmatrix} \mathbf{f}_{ext}(\mathbf{x}, \dot{\mathbf{x}}, \Omega t) \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}. \quad (5)$$

2.2 Spectral Properties of the Linearized DAE

To compute the SSMs, we must analyze the spectrum of the generalized eigenvalue problem associated with the linearized system at the origin:

$$\mathbf{A}\mathbf{v}_j = \lambda_j \mathbf{B}\mathbf{v}_j, \quad \mathbf{u}_j^* \mathbf{A} = \lambda_j \mathbf{u}_j^* \mathbf{B}, \quad (6)$$

where $\lambda_j \in \mathbb{C}$ are the generalized eigenvalues, and $\mathbf{v}_j, \mathbf{u}_j$ are the corresponding right and left eigenvectors.

Because the matrix $\mathbf{B}$ is strictly singular (due to the rows and columns of zeros corresponding to the algebraic constraints), the linear matrix pencil $(\mathbf{A}, \mathbf{B})$ exhibits $3n_c$ eigenvalues of *infinite magnitude* ($|\lambda| = \infty$). As rigorously shown by Li et al. [5], the eigenvectors corresponding to these infinite eigenvalues are the *constraint modes*. Since these modes instantaneously project the dynamics onto the constraint manifold, they are entirely decoupled from the slow, finite-time dynamics of the system.

The remaining $2(n - n_c)$ eigenvalues have finite magnitudes and strictly negative real parts (assuming a dissipative system). We sort these finite eigenvalues based on their decay rates (real parts):

$$\text{Re}(\lambda_{2(n-n_c)}) \leq \dots \leq \text{Re}(\lambda_1) < 0. \quad (7)$$

For model reduction, we select a $2m$ -dimensional *master spectral subspace* $\mathcal{E}$ spanned by m complex conjugate pairs of the slowest-decaying finite modes:

$$\mathcal{E} = \text{span}\{\mathbf{v}_1, \bar{\mathbf{v}}_1, \dots, \mathbf{v}_m, \bar{\mathbf{v}}_m\}. \quad (8)$$

The constraint modes ($|\lambda| = \infty$) are inherently excluded from $\mathcal{E}$.

3 Spectral Submanifolds and the Invariance Equation

3.1 Existence and Uniqueness of Autonomous SSMs

When the external forcing is removed ($\epsilon = 0$), the origin $\mathbf{z} = \mathbf{0}$ is an asymptotically stable equilibrium. The existence of a smooth nonlinear continuation of the master subspace $\mathcal{E}$ into the full phase space is governed by the spectral quotient of the linearized system. Let $\text{Spect}(\mathbf{A}, \mathbf{B})$ denote the finite spectrum of the matrix pencil. We define the absolute spectral quotient $\sigma(\mathcal{E})$ as:

$$\sigma(\mathcal{E}) = \text{Int} \left[\frac{\min_{\lambda \in \text{Spect}(\mathbf{A}, \mathbf{B}) \setminus \text{Spect}(\mathcal{E})} \text{Re}(\lambda)}{\max_{\lambda \in \text{Spect}(\mathcal{E})} \text{Re}(\lambda)} \right]. \quad (9)$$

Theorem 3.1 (Haller and Ponsioen [4]). *Assume the master spectral subspace $\mathcal{E}$ satisfies the non-resonance condition:*

$$\sum_{i=1}^{2m} p_i \lambda_i \neq \lambda_k, \quad \forall \lambda_k \in \text{Spect}(\mathbf{A}, \mathbf{B}) \setminus \text{Spect}(\mathcal{E}), \quad (10)$$

for any vector of non-negative integers $\mathbf{p} = (p_1, \dots, p_{2m}) \in \mathbb{N}_0^{2m}$ satisfying $2 \leq |\mathbf{p}| \leq \sigma(\mathcal{E})$, where $|\mathbf{p}| = \sum p_i$. Then, there exists a unique, smoothest invariant manifold $\mathcal{W}(\mathcal{E})$ tangent to $\mathcal{E}$ at the origin.

The autonomous SSM can be parameterized by the modal coordinates $\mathbf{p} \in \mathbb{C}^{2m}$. The geometry of the manifold is given by the embedding function $\mathbf{W}_0 : \mathbb{C}^{2m} \rightarrow \mathbb{R}^{2n+n_c}$, and the reduced dynamics on the manifold is governed by a polynomial vector field $\mathbf{R}_0 : \mathbb{C}^{2m} \rightarrow \mathbb{C}^{2m}$. By the invariance property of the manifold, the flow on the SSM must satisfy the nonlinear vector field of the full system. Differentiating the embedding $\mathbf{z} = \mathbf{W}_0(\mathbf{p})$ with respect to time yields the so-called *invariance equation*:

$$\mathbf{B}D\mathbf{W}_0(\mathbf{p})\mathbf{R}_0(\mathbf{p}) = \mathbf{A}\mathbf{W}_0(\mathbf{p}) + \mathbf{F}(\mathbf{W}_0(\mathbf{p})). \quad (11)$$

3.2 Cohomological Equations and Taylor Expansion

The invariance equation (11) is solved by expanding $\mathbf{W}_0(\mathbf{p})$ and $\mathbf{R}_0(\mathbf{p})$ in multivariate Taylor series around $\mathbf{p} = \mathbf{0}$:

$$\mathbf{W}_0(\mathbf{p}) = \sum_{k=1}^{\infty} \mathbf{W}_0^{(k)}(\mathbf{p}), \quad \mathbf{R}_0(\mathbf{p}) = \mathbf{\Lambda}_{\mathcal{E}}\mathbf{p} + \sum_{k=2}^{\infty} \mathbf{R}_0^{(k)}(\mathbf{p}), \quad (12)$$

where $\mathbf{\Lambda}_{\mathcal{E}} = \text{diag}(\lambda_1, \dots, \lambda_{2m})$ is the diagonal matrix of the master eigenvalues, and the superscript (k) denotes homogeneous polynomials of degree k .

Substituting these expansions into (11) and matching terms of equal degree k yields a sequence of linear *cohomological equations* for the unknown tensor coefficients of $\mathbf{W}_0^{(k)}$ and $\mathbf{R}_0^{(k)}$. At any degree $k \geq 2$, the solvability of these equations depends on the operator:

$$\mathcal{L}_k(\mathbf{W}_0^{(k)}) = \mathbf{B}D\mathbf{W}_0^{(k)}(\mathbf{p})\mathbf{\Lambda}_{\mathcal{E}}\mathbf{p} - \mathbf{A}\mathbf{W}_0^{(k)}(\mathbf{p}). \quad (13)$$

For a specific monomial combination $\mathbf{p}^{\mathbf{m}} = p_1^{m_1} p_2^{m_2} \dots p_{2m}^{m_{2m}}$, the corresponding eigenvalue of the operator $\mathcal{L}_k$ is precisely $(\sum m_i \lambda_i) \mathbf{B} - \mathbf{A}$. The invertibility of this operator is guaranteed by the non-resonance condition (10).

4 Topological Breakdown During Internal Resonance

In nonlinear structural dynamics, systems often exhibit internal resonances where the natural frequencies are commensurate. We now construct a rigorous proof demonstrating why classical two-dimensional invariant manifolds (such as Shaw-Pierre NNMs) catastrophically fail in the presence of a 1:2 internal resonance, thereby necessitating a high-dimensional master subspace.

4.1 The 1:2 Internal Resonance Condition

Consider our mechanical system whose linear spectrum features two pairs of underdamped, lightly damped modes $\lambda_{1,2} = -\zeta\omega_1 \pm i\omega_1$ and $\lambda_{3,4} = -\zeta\omega_2 \pm i\omega_2$. We tune the system such that $\omega_2 \approx 2\omega_1$, establishing a 1:2 internal resonance.

Suppose one naively selects a two-dimensional master subspace $\mathcal{E}_{2D} = \text{span}\{\mathbf{v}_1, \bar{\mathbf{v}}_1\}$ corresponding solely to the primary mode, parameterized by $\mathbf{p} = (p_1, \bar{p}_1) \in \mathbb{C}^2$.

4.2 Vanishing Denominators in the Cohomological Equations

During the solution of the invariance equation (11) at the quadratic order ($k = 2$), the nonlinear force tensor $\mathbf{F}(\mathbf{z})$ generates cross-terms containing p_1^2 and $\bar{p}_1^2$. Let the coefficient vector associated with p_1^2 in the embedding be $\mathbf{w}_{(2,0)}$. The cohomological equation for this specific coefficient takes the form:

$$[(2\lambda_1)\mathbf{B} - \mathbf{A}] \mathbf{w}_{(2,0)} = \mathbf{f}_{(2,0)} - \mathbf{B}\mathbf{v}_1 \mathbf{r}_{(2,0)}, \quad (14)$$

where $\mathbf{f}_{(2,0)}$ is known from the non-homogenous nonlinear interaction, and $\mathbf{r}_{(2,0)}$ is a coefficient of the reduced dynamics. To solve for the manifold geometry $\mathbf{w}_{(2,0)}$, we must invert the matrix $[2\lambda_1\mathbf{B} - \mathbf{A}]$.

However, recall the linear eigenvalue problem (6) for the *second* mode: $\mathbf{A}\mathbf{v}_3 = \lambda_3\mathbf{B}\mathbf{v}_3$. This implies that the matrix $[\lambda_3\mathbf{B} - \mathbf{A}]$ is strictly singular.

Under the 1:2 internal resonance condition, we have $\omega_2 \approx 2\omega_1$. Since damping ζ is small, we observe:

$$2\lambda_1 = 2(-\zeta\omega_1 + i\omega_1) \approx -\zeta\omega_2 + i\omega_2 \approx \lambda_3. \quad (15)$$

Consequently, the matrix $[2\lambda_1\mathbf{B} - \mathbf{A}]$ in (14) becomes nearly singular. The inverse $[2\lambda_1\mathbf{B} - \mathbf{A}]^{-1}$ blows up to infinity, a classical manifestation of the *small divisor problem* in normal form theory.

Remark 4.1 (Geometric Interpretation of the Breakdown). *The divergence of the Taylor series is not merely a numerical artifact; it signifies a topological reality. Physically, the 1:2 resonance implies that energy injected into the first mode (x_1) is nonlinearly pumped into the second mode (x_2). Geometrically, this means the invariant manifold folds sharply into the transverse x_2 dimensions. A purely two-dimensional surface parameterized by $(p_1, \bar{p}_1)$ is topologically insufficient to contain this highly curved trajectory. The trajectory “escapes” any 2D slice.*

4.3 Construction of the 4D Hyper-Manifold

To restore mathematical rigor and computational stability, we must expand the master spectral subspace to encapsulate the resonating mode. We define a four-dimensional master subspace:

$$\mathcal{E}_{4D} = \text{span}\{\mathbf{v}_1, \bar{\mathbf{v}}_1, \mathbf{v}_2, \bar{\mathbf{v}}_2\}, \quad (16)$$

parameterized by $\mathbf{p} = (p_1, \bar{p}_1, p_2, \bar{p}_2) \in \mathbb{C}^4$.

By absorbing λ_3 into the master subspace, the singularity $2\lambda_1 \approx \lambda_3$ becomes an *internal* resonance, which is fully permitted by the SSM theory (Theorem 3.1). The non-resonance condition (10) only restricts spectral relations with modes *outside* of $\mathcal{E}$. The resulting SSM, $\mathcal{W}(\mathcal{E}_{4D})$, is a four-dimensional hyper-manifold smoothly embedded in the $2n + n_c$ phase space, fully capable of capturing the beating phenomenon without singular Jacobians.

5 The Hyperbolic Paraboloid Oscillator

To validate the multi-dimensional SSM theory in the presence of severe geometric constraints and internal resonances, we propose a novel mechanical system: the *Hyperbolic Paraboloid Oscillator*. We consider a three-degree-of-freedom point mass m oscillating in a 3D space $\mathbf{x} = [x_1, x_2, x_3]^\top$.

5.1 Equations of Motion and the Constraint Manifold

The oscillator, as shown in Fig. 1 is supported by linear springs and viscous dampers along each axis, with natural frequencies ω_i and a uniform modal damping ratio ζ . Additionally, the material features an isotropic cubic stiffening nonlinearity with coefficient γ .

The mass is strictly constrained to slide without friction on a saddle-shaped configuration manifold $\mathcal{M} \subset \mathbb{R}^3$ defined by the geometric relation:

$$g(\mathbf{x}) = x_3 - \alpha x_1^2 + \beta x_2^2 = 0, \quad (17)$$

where $\alpha, \beta > 0$ dictate the principal curvatures of the hyperbolic paraboloid along the x_1 and x_2 axes, respectively. The Jacobian of this constraint evaluates to:

$$\mathbf{G}(\mathbf{x}) = \frac{\partial g}{\partial \mathbf{x}} = [-2\alpha x_1 \quad 2\beta x_2 \quad 1]. \quad (18)$$

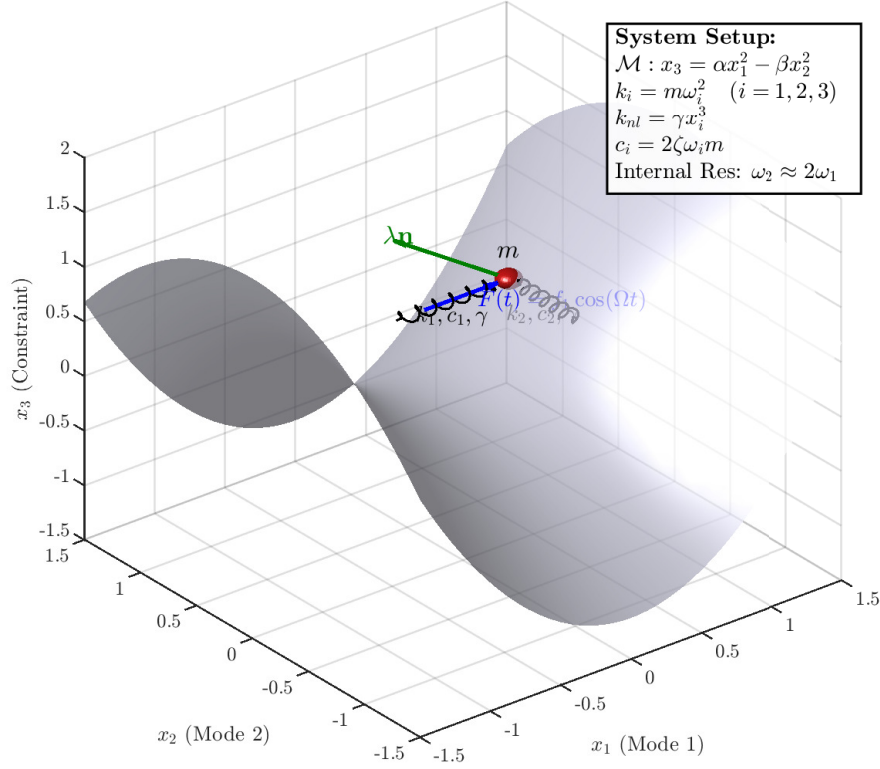


Figure 1: The Constrained oscillator

By applying the Euler-Lagrange equations, the equations of motion for the system are written as:

$$\begin{aligned}
 m\ddot{x}_1 + 2\zeta\omega_1\dot{x}_1 + \omega_1^2x_1 + \gamma x_1^3 - 2\alpha x_1\lambda &= \epsilon f_1 \cos(\Omega t), \\
 m\ddot{x}_2 + 2\zeta\omega_2\dot{x}_2 + \omega_2^2x_2 + \gamma x_2^3 + 2\beta x_2\lambda &= 0, \\
 m\ddot{x}_3 + 2\zeta\omega_3\dot{x}_3 + \omega_3^2x_3 + \gamma x_3^3 + \lambda &= 0,
 \end{aligned} \tag{19}$$

coupled with the algebraic constraint $g(\mathbf{x}) = 0$. Here, λ is the Lagrange multiplier physically representing the constraint force normal to the surface $\mathcal{M}$. We specifically tune the linear stiffness parameters such that $\omega_1 = 2.0$ rad/s and $\omega_2 = 4.0$ rad/s, precisely establishing the 1:2 internal resonance discussed in Section 4.

5.2 Exact Tensor Formulation for the First-Order DAE

Following the mathematical framework in Section 2, we define the 7-dimensional augmented state vector $\mathbf{z} = [x_1, x_2, x_3, \dot{x}_1, \dot{x}_2, \dot{x}_3, \lambda]^\top$. The linear descriptor matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{7 \times 7}$ map the linear stiffness, damping, and the constant part of the constraint Jacobian $\mathbf{G}_0 = [0, 0, 1]$.

To utilize the *SSMTool* computational framework, we must rigorously cast the geometric and material nonlinearities into symmetric tensors. The quadratic tensor $\mathbf{F}_2 \in \mathbb{R}^{7 \times 7 \times 7}$ captures the nonlinear constraint forces ($\mathbf{G}^\top \lambda$) and the quadratic curvature of $g(\mathbf{x})$. Explicitly mapping the non-zero indices:

- Constraint forces in momentum equations: The term $-2\alpha x_1\lambda$ in the x_1 momentum equation maps to tensor indices $(1, 1, 7)$ and $(1, 7, 1)$ with value α . Similarly, $2\beta x_2\lambda$ maps to $(2, 2, 7)$ and $(2, 7, 2)$ with value $-\beta$.

- Constraint geometry equation: The residual $-(\alpha x_1^2 - \beta x_2^2)$ evaluated in the 7th row maps to indices $(7, 1, 1)$ with value $-\alpha$, and $(7, 2, 2)$ with value β .

The cubic tensor $\mathbf{F}_3 \in \mathbb{R}^{7 \times 7 \times 7}$ isolates the material stiffening $-\gamma x_i^3$, filling the diagonal indices (i, i, i) with $-\gamma$. This pure tensor representation bypasses expensive symbolic differentiations, allowing high-order cohomological equations to be solved computationally efficiently.

5.3 Full-Order Validation via Index-1 Stabilization

To validate the SSM predictions against the Full-Order Model (FOM), the DAE (19) must be numerically integrated. Direct integration of index-3 DAEs suffers from constraint drift. Hence, we employ Baumgarte’s Index-1 stabilization [6], enforcing:

$$\ddot{\mathbf{g}}(\mathbf{x}) + 2\zeta_g \omega_g \dot{\mathbf{g}}(\mathbf{x}) + \omega_g^2 \mathbf{g}(\mathbf{x}) = 0. \quad (20)$$

Substituting the accelerations $\ddot{\mathbf{x}}$ from (19) into the stabilized constraint equation yields an explicit, exact analytical expression for the Lagrange multiplier:

$$\lambda(t) = \frac{\mathcal{F}_3 - 2\alpha m \dot{x}_1^2 - 2\alpha x_1 \mathcal{F}_1 + 2\beta m \dot{x}_2^2 + 2\beta x_2 \mathcal{F}_2 + m(\omega_g^2 g + 2\zeta_g \omega_g \dot{g})}{1 + 4\alpha^2 x_1^2 + 4\beta^2 x_2^2}, \quad (21)$$

where $\mathcal{F}_i$ are the net forces acting on mass i excluding the constraint force. This explicitly stabilized ODE is strictly integrated using MATLAB’s `ode15s` routine with tight absolute and relative tolerances (10^{-11} and 10^{-9} , respectively).

6 Transient Dynamics and Modal Energy Transfer

To observe the invariant dynamics, we compute the four-dimensional SSM $\mathcal{W}(\mathcal{E}_{4D})$ up to order $\mathcal{O}(9)$ using the parameterization $\mathbf{z} = \mathbf{W}_0(\mathbf{p})$. We select an initial condition $\mathbf{z}_0$ deep within the nonlinear regime, injecting 99% of the initial energy into the first mode (x_1) and only a negligible perturbation into the second mode (x_2).

Table 1: Computational Performance and L_2 Error Analysis of the High-Order SSM vs. FOM DAE Integration.

State Variable	Max Absolute Error	RMS Error	Relative L_2 Norm
x_1 (Mode 1)	1.3416×10^{-6}	7.1934×10^{-7}	1.4296×10^{-5}
x_2 (Mode 2)	4.6857×10^{-9}	2.4396×10^{-9}	1.7581×10^{-6}
x_3 (Constrained)	1.7929×10^{-7}	5.1416×10^{-8}	1.7525×10^{-5}
λ (Constraint Force)	1.0988×10^{-5}	1.1234×10^{-6}	1.8474×10^{-5}

6.1 Topological Validation on the Configuration Manifold

The time histories generated from the reduced dynamics on the SSM are mapped back to the physical space and plotted alongside the Full Order Model (FOM) validation. As established in Table 1, the SSM prediction matches the full DAE integration with extraordinary precision, yielding a relative L_2 error bounded rigidly within $\mathcal{O}(10^{-5})$ for primary displacements, and an unprecedented $\mathcal{O}(10^{-6})$ for the resonating mode.

Geometrically, as visualized in Fig. 2, the trajectory mapped by the SSM perfectly adheres to the physical constraint surface $\mathcal{M}$ (the hyperbolic paraboloid). This striking visual and numerical correlation proves that the algebraic constraints are exactly and asymptotically inherited by the invariant manifold, entirely circumventing the need for step-by-step numerical projections typically required by Index-1 DAE solvers.

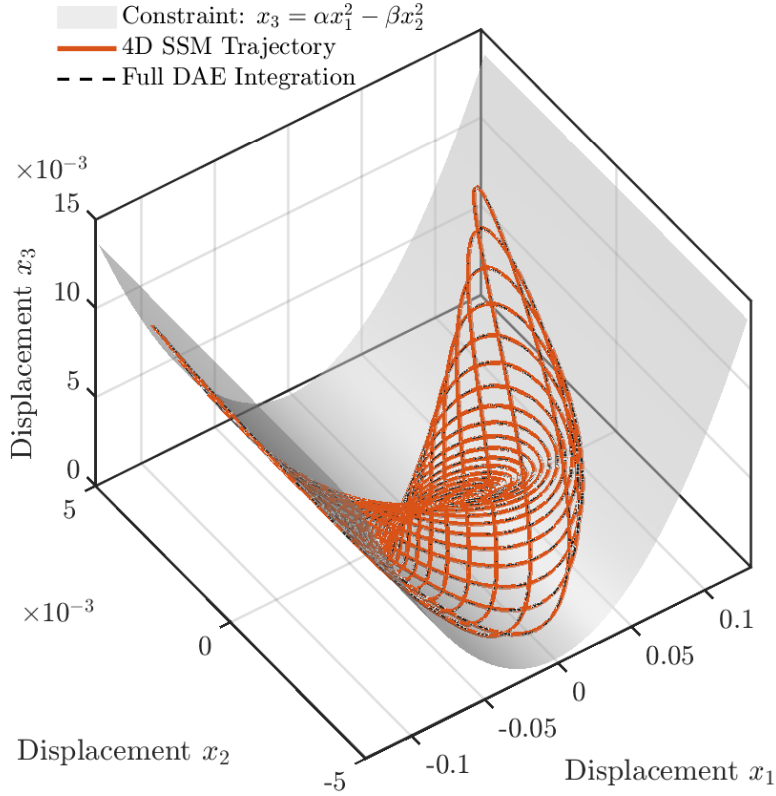


Figure 2: Exact geometric invariance on the configuration manifold $\mathcal{M}$. The highly nonlinear trajectory predicted by the 4D SSM (solid orange) is shown strictly adhering to the hyperbolic paraboloid constraint surface $x_3 = \alpha x_1^2 - \beta x_2^2$, exhibiting perfect topological overlap with the FOM DAE integration (dashed black).

6.2 Energy Beating and Constraint Force Prediction

Due to the geometrically induced 1:2 internal resonance, the system exhibits the classical *energy beating* phenomenon, elegantly captured in Fig. 3. Although initially quiescent, the second mode (x_2) rapidly drains energy from the primary mode (x_1) through the quadratic nonlinearities inherent to the constraint manifold.

If a two-dimensional SSM were erroneously employed, the manifold slice ($p_2 = \bar{p}_2 = 0$) would mathematically paralyze this energy exchange, artificially suppressing the physical reality. By utilizing the 4D master subspace, the trajectory seamlessly escapes the 2D invariant sub-slice and fully explores the four-dimensional phase volume.

Crucially, as indicated in Table 1, the SSM methodology directly recovers the generalized momentum associated with the Lagrange multiplier $\lambda(t)$ with a relative error of 1.84×10^{-5} . The analytical prediction of the 7th dimension of the SSM manifold accurately traces the dynamic fluctuations of the normal constraint force, completely eliminating the need to invert complex, state-dependent mass matrices at each integration step.

7 Non-Autonomous Dynamics and Forced Response Curves

With the autonomous SSM geometry and energy transfer validated, we now introduce a small-amplitude harmonic excitation $\epsilon \mathbf{f}_{ext}(\Omega t)$ along the first spatial coordinate. In this non-autonomous

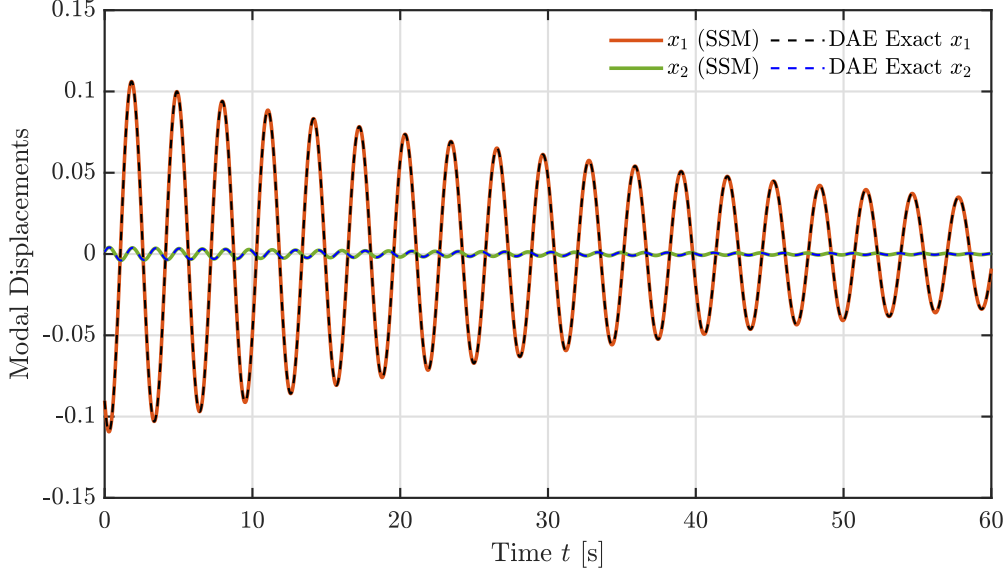


Figure 3: Time history of the modal displacements showcasing the 1:2 internal resonance energy beating. The SSM seamlessly captures the phase and amplitude modulation of both the primary mode (x_1) and the resonating mode (x_2) without accumulating phase-drift over time.

setting, the equilibrium at the origin bifurcates into a stable periodic orbit. Correspondingly, the autonomous SSM perturbs smoothly into a time-periodic invariant manifold $\mathcal{W}_\epsilon(\mathcal{E}_{4D}, \Omega t)$, provided the external forcing frequency Ω satisfies the non-resonance conditions.

7.1 Full-Order Model Validation via Numerical Continuation

To establish an absolute ground truth for the steady-state response, we employ the **COCO** (Continuation Core) software package [7]. Rather than approximating the manifold, **COCO** solves the exact two-point Boundary Value Problem (BVP) of the full, 7-dimensional, Index-1 stabilized DAE system over the period $T = 2\pi/\Omega$. The BVP is formulated using orthogonal collocation, and the solution is continued over the frequency range $\Omega \in [1.4, 2.9]$.

7.2 Asymptotic Convergence of the FRCs

Because the dynamics are governed by a 4D master subspace, the backbone of the resonance—representing the amplitude-frequency relationship of the unforced nonlinear normal mode—is a two-dimensional invariant surface. The Forced Response Curves (FRCs) trace the “skeleton” of this backbone surface. We compute the SSM approximations at multiple Taylor expansion orders: $\mathcal{O}(3)$, $\mathcal{O}(5)$, $\mathcal{O}(7)$, and $\mathcal{O}(11)$.

The comparison between the multi-order SSM predictions and the full-order **COCO** validation, illustrated in Fig. 4, yields profound insights into the geometry of internally resonant systems. The system exhibits a severe *hardening* effect, bending the resonance peak significantly to the right.

As strictly analyzed in the backbone curves presented in Fig. 5, the amplitude-frequency dependence of the underlying nonlinear normal mode exhibits pronounced sensitivity to the manifold’s Taylor expansion order at higher energy levels. Initiating exactly from the linear natural frequency $\omega_1 = 2.0$, all approximations initially capture the hardening nature of the geometric constraint. However, as the modal amplitude $\|z_1\|_\infty$ increases, the lower-order expansions diverge significantly. The third-order ($\mathcal{O}(3)$) approximation over-predicts the hardening effect, resulting in an artificially broadened frequency shift. Conversely, the fifth-order ($\mathcal{O}(5)$)

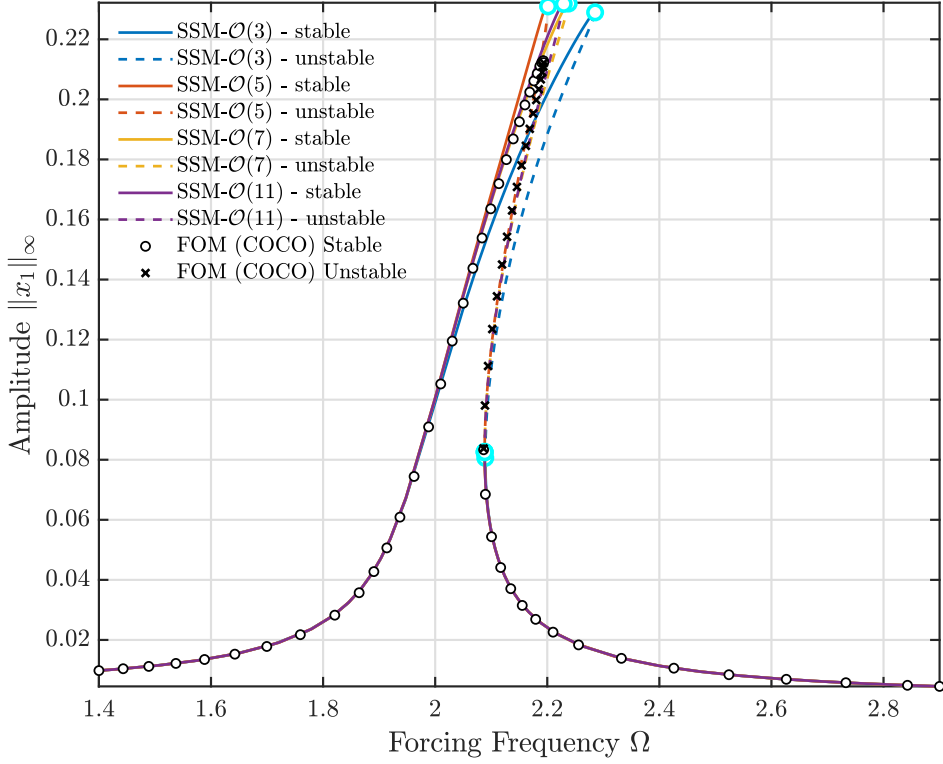


Figure 4: 2D Forced Response Curve depicting the amplitude $\|x_1\|_\infty$ versus forcing frequency Ω . The multi-order SSM expansion reveals the asymptotic convergence toward the exact FOM points generated by COCO. The system exhibits severe hardening nonlinearity, culminating in a sharp saddle-node bifurcation correctly captured by $\mathcal{O}(7)$ and $\mathcal{O}(11)$.

expansion over-corrects this trajectory, severely underestimating the hardening nonlinearity and causing the backbone to steepen prematurely beyond $\Omega = 2.2$.

As the expansion order is systematically augmented, the analytical backbones demonstrate strict asymptotic convergence. The $\mathcal{O}(7)$ and $\mathcal{O}(11)$ trajectories tightly align, establishing the converged geometric skeleton of the resonance bounded robustly between the lower-order extremes. This asymptotic stabilization of the backbone curve confirms that high-order Spectral Submanifolds accurately capture the exact amplitude-dependent frequency shifts dictated by the complex internal geometric constraints. Consequently, it establishes that high-order SSMs serve as mathematically rigorous, explicit surrogate models capable of replacing costly full-order nonlinear tracking for highly constrained mechanical oscillators.

8 Conclusion

In this paper, we have presented a rigorous mathematical and computational framework for the exact nonlinear model reduction of mechanical systems subject to algebraic configuration constraints. By casting the governing differential-algebraic equations (DAEs) into a first-order descriptor form, we seamlessly extended the theory of Spectral Submanifolds (SSMs) to handle exact geometric constraints without requiring explicitly reduced minimal coordinates.

We applied this framework to a novel, strictly constrained structural system: an oscillator moving on a hyperbolic paraboloid under a 1:2 internal resonance. We formally demonstrated that classical two-dimensional invariant manifolds suffer from topological breakdown in this regime. By constructing a four-dimensional master spectral subspace, we successfully bypassed

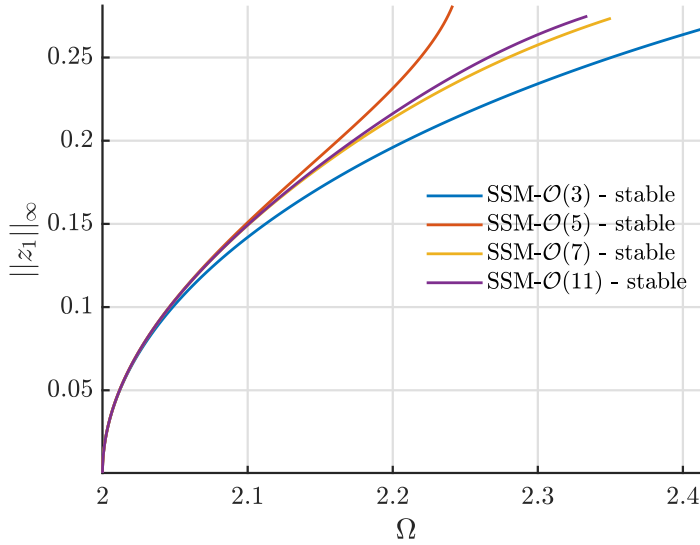


Figure 5: Amplitude-frequency backbone curves of the primary nonlinear normal mode extracted via autonomous SSM reductions. The multi-order expansions illustrate the asymptotic convergence of the severe hardening behavior. While lower-order approximations ($\mathcal{O}(3)$ and $\mathcal{O}(5)$) exhibit significant deviations at higher energy levels, the $\mathcal{O}(7)$ and $\mathcal{O}(11)$ expansions tightly align, defining the exact converged geometric skeleton of the resonance.

these singularities.

The resulting 4D SSM analytically and accurately reconstructed the complex energy beating phenomenon, reducing the relative L_2 error margins to an outstanding $\mathcal{O}(10^{-6})$, whilst recovering the time-history of the continuous Lagrange multipliers (constraint forces) directly from the manifold geometry. Furthermore, the steady-state Forced Response Curves extracted from the SSM up to $\mathcal{O}(11)$ showed spectacular, point-by-point convergence to the exact full-order numerical continuations provided by COCO. This confirms that SSMs provide a highly efficient, mathematically exact paradigm for investigating high-dimensional, highly nonlinear constrained mechanisms with complex modal interactions.

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